

# MITS SIOA-REV-2 Technical Reference

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## SIOA-REV-2 Technical Reference

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### Overview

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The SIOA-REV-2 (Serial I/O Adapter, Revision 2) is an Altair bus peripheral card designed for MITS Altair 8800 and compatible systems. It provides one full-duplex RS-232 serial port with:

- Programmable baud rates from 110 to 19,200 baud (via 8-position DIP switch)
- Programmable word format: 5–8 data bits, 1–2 stop bits, odd/even/no parity (via 5-position DIP switch)
- Configurable I/O port address (via 7-position DIP switch)
- Interrupt-driven operation with selectable interrupt vector (VI0–VI7)
- RS-232 DCE and DTE connectors (DB25 and DE9) plus an FTDI-compatible 6-pin TTL header
- On-board +5V regulation from the Altair +8V supply, plus –12V generation from the Altair –16V supply for the UART negative rail

The board interfaces to the Altair (S-100) bus through the 100-pin edge connector. The CPU (8080 or Z80) communicates with the card using two I/O port addresses: one for data and one for status/control.

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### Project Resources

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The SIOA-REV-2 project is hosted on GitHub at <https://github.com/deltecent/88-SIOR2>.

There you will find:

- **Test programs** — software for verifying board operation
- **Wiki** — additional documentation and usage notes
- **Issue tracker** — report bugs or request improvements

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### IC Descriptions

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#### U1 — MAX232 (RS-232 Level Converter, DIP-16)

**Function:** Converts between TTL/CMOS logic levels (0/5V) and RS-232 voltage levels ( $\pm 10V$ ). Requires only a +5V supply; internal charge pumps generate  $\pm 10V$  for RS-232 drivers.

Contains two RS-232 line drivers and two RS-232 line receivers.

| Pin(s) | Name     | Direction | Description                                                   |
|--------|----------|-----------|---------------------------------------------------------------|
| 1, 3   | C1+, C1– | —         | External charge pump capacitor (1 $\mu$ F) for +10V generator |
| 4, 5   | C2+, C2– | —         | External charge pump capacitor (1 $\mu$ F) for –10V generator |

| Pin(s) | Name         | Direction | Description                                    |
|--------|--------------|-----------|------------------------------------------------|
| 2      | V+           | Out       | +10V internal rail                             |
| 6      | V-           | Out       | -10V internal rail                             |
| 11, 10 | T1IN, T2IN   | In        | TTL inputs to RS-232 drivers                   |
| 14, 7  | T1OUT, T2OUT | Out       | RS-232 outputs (+/-10V), connected to TXD line |
| 13, 8  | R1IN, R2IN   | In        | RS-232 inputs (from RXD line)                  |
| 12, 9  | R1OUT, R2OUT | Out       | TTL outputs from RS-232 receivers              |
| 15     | GND          | —         | Ground                                         |
| 16     | VCC          | —         | +5V supply                                     |

### In this design:

- T1IN receives TSO (Transmitter Serial Output) from U2 UART
- T1OUT drives TXD line to RS-232 connectors (J2, J3, J4, J5)
- R1IN receives RXD from RS-232 connectors
- R1OUT drives RSI (Receiver Serial Input) to U2 UART

**Important — TTL header use:** U1's R1OUT drives the same RSI net that the J1 TTL header exposes. **U1 must be removed from its socket when using the J1 TTL header** (e.g. with an FTDI USB-serial adapter), otherwise U1's receiver output contends with the external device driving RSI. With U1 removed, RS-232 operation on J2–J5 is disabled and only the TTL header is active.

## U2 — COM2502 (UART, DIP-40)

**Function:** Universal Asynchronous Receiver/Transmitter (AY-5-1013 family). Performs parallel-to-serial conversion for transmission and serial-to-parallel conversion for reception. Includes programmable word format (data bits, stop bits, parity), status flags, and tristate data outputs. The COM2502 requires a dual supply: +5V on VCC (pin 1) and **-12V on VGG (pin 2)**. The -12V rail is generated on-board from the Altair -16V supply (see Power Supply). The pin-compatible CMOS Intersil IM6402 runs from +5V only and leaves pin 2 unconnected.

The UART operates at 16× the desired baud rate on its clock inputs.

### Compatible UARTs

This socket accepts any of the standard 40-pin AY-5-1013-pinout UARTs. The only wiring difference between them is the supply on pin 2 (VGG): the older PMOS parts require -12V there, while the later +5V-only (NMOS/CMOS) parts leave pin 2 unconnected. Install jumper **JP2** to enable the on-board -12V supply only for the parts that need it; leave JP2 open for the +5V-only parts.

| UART                                     | -12V jumper (JP2)       |
|------------------------------------------|-------------------------|
| General Instrument AY-5-1012, AY-5-1013A | Required                |
| Texas Instruments TMS6011                | Required                |
| AMI S1883                                | Required                |
| SMC COM2502, COM2502H, COM2017           | Required                |
| SMC COM8502, COM8017                     | Not required (+5V only) |

| UART                           | -12V jumper (JP2)       |
|--------------------------------|-------------------------|
| Western Digital TR1402, TR1602 | Required                |
| Western Digital TR1863, TR1865 | Not required (+5V only) |
| Intersil IM6402                | Not required (+5V only) |

None of these parts require +12V — the dual-supply (PMOS) parts use +5V and -12V only, which is exactly what the board provides. On the +5V-only parts, pin 2 (VGG) is a no-connect, so leave JP2 open to avoid feeding -12V to an unused pin.

## Pin Summary

| Pin  | Name      | Dir | Description                                                                                                                                                              |
|------|-----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | VCC       | In  | +5V supply                                                                                                                                                               |
| 2    | VGG       | In  | -12V supply (from on-board -16V→-12V regulator)                                                                                                                          |
| 3    | GND       | In  | Ground                                                                                                                                                                   |
| 4    | /RDE      | In  | Receiver Data Enable (active low): enables RBR outputs onto bDI bus                                                                                                      |
| 5–12 | RBR8–RBR1 | Out | Receiver Buffer Register, bits 8–1 (MSB first). Tristate; enabled by /RDE low. Also serve as status word outputs when /SWE low                                           |
| 13   | PE        | Out | Parity Error (high = parity error detected). Tristate; enabled when /SWE (pin 16) is low; connected to bDI2                                                              |
| 14   | FE        | Out | Framing Error (high = invalid stop bit). Tristate; enabled when /SWE (pin 16) is low; connected to bDI3                                                                  |
| 15   | OVR       | Out | Overflow Error (high = previous character not read). Tristate; enabled when /SWE (pin 16) is low; connected to bDI4                                                      |
| 16   | /SWE      | In  | Status Word Enable (active low): enables all status outputs — RBR pins 5–12, PE, FE, OVR pins 13–15, DR, and TBMT. Connected to RDE; goes low on any non-data-read cycle |
| 17   | RC        | In  | Receiver Clock input (16× baud rate)                                                                                                                                     |
| 18   | /RDAV     | In  | Receiver Data Available (tied to DR, active low, clears DR flag when read)                                                                                               |
| 19   | DR        | Out | Data Ready (high = received character is in RBR, ready to read)                                                                                                          |
| 20   | RSI       | In  | Receiver Serial Input (TTL-level serial data in)                                                                                                                         |
| 21   | MR        | In  | Master Reset (active high; clears all status, disables outputs)                                                                                                          |
| 22   | TBMT      | Out | Transmitter Buffer Empty (high = TBR can accept new character)                                                                                                           |
| 23   | /TDS      | In  | Transmitter Data Strobe (active low pulse: loads TBR from TBR1–TBR8)                                                                                                     |
| 24   | TRE       | Out | Transmitter Register Empty (high = shift register is empty)                                                                                                              |
| 25   | TSO       | Out | Transmitter Serial Output (serial data output)                                                                                                                           |

| Pin   | Name      | Dir | Description                                                                               |
|-------|-----------|-----|-------------------------------------------------------------------------------------------|
| 26–33 | TBR1–TBR8 | In  | Transmitter Buffer Register inputs, bits 1–8 (LSB first)                                  |
| 34    | CS        | In  | Control Strobe (active high pulse: loads NDB1, NDB2, NSB, NPB, POE into control register) |
| 35    | NPB       | In  | No Parity Bit (high = no parity bit transmitted/checked)                                  |
| 36    | NSB       | In  | Number of Stop Bits (0 = 1 stop bit; 1 = 2 stop bits, or 1.5 for 5-bit words)             |
| 37    | NDB2      | In  | Number of Data Bits bit 2 (MSB of word length select)                                     |
| 38    | NDB1      | In  | Number of Data Bits bit 1 (LSB of word length select)                                     |
| 39    | POE       | In  | Parity Odd/Even select (0 = even parity; 1 = odd parity)                                  |
| 40    | TC        | In  | Transmitter Clock input (16× baud rate)                                                   |

### Word Length Truth Table (NDB1, NDB2)

| NDB2 | NDB1 | Data Bits |
|------|------|-----------|
| 0    | 0    | 5 bits    |
| 0    | 1    | 6 bits    |
| 1    | 0    | 7 bits    |
| 1    | 1    | 8 bits    |

### Stop Bit Truth Table (NSB)

| NSB | Stop Bits                                                  |
|-----|------------------------------------------------------------|
| 0   | 1 stop bit                                                 |
| 1   | 2 stop bits (1.5 stop bits when word length = 5 data bits) |

### Parity Truth Table (NPB, POE)

NPB selects whether a parity bit is generated/checked; POE selects odd vs. even and is a don't-care when parity is disabled.

| NPB | POE | Parity                   |
|-----|-----|--------------------------|
| 1   | X   | No parity bit (disabled) |
| 0   | 0   | Even parity              |
| 0   | 1   | Odd parity               |

### Status Bit Summary

| Signal | Pin | Meaning when High                                                  |
|--------|-----|--------------------------------------------------------------------|
| DR     | 19  | Data Ready: received character is available in RBR                 |
| TBMT   | 22  | TX Buffer Empty: transmitter is ready to accept a new character    |
| TRE    | 24  | TX Register Empty: transmitter shift register has finished sending |

| Signal | Pin | Meaning when High                                                         |
|--------|-----|---------------------------------------------------------------------------|
| PE     | 13  | Parity Error: last received character had a parity error                  |
| FE     | 14  | Framing Error: last received character had an invalid stop bit            |
| OVR    | 15  | Overrun: received character lost (DR was not cleared before next arrived) |

## U7 — 74LS08 (Quad 2-Input AND Gate, DIP-14)

**Function:** Four independent 2-input AND gates. Used to combine the two port-decoded read/write strobes into unified bus-enable signals.

### Truth Table (each gate)

| A | B | Y = A AND B |
|---|---|-------------|
| 0 | 0 | 0           |
| 0 | 1 | 0           |
| 1 | 0 | 0           |
| 1 | 1 | 1           |

**Gate A:** /DIEN = /SWE AND /RDE - Goes LOW (active) when either a status read or data read is occurring - Enables U13 (74LS244) to drive the Altair DI bus during any read cycle

**Gates B, C and D:** Unused; inputs tied off, outputs unconnected. (U14, the DO-bus buffer, is permanently enabled and no longer requires a /DOEN strobe.)

## U15 — 74LS368 (Hex Inverting Tristate Buffer, DIP-16)

**Function:** Six independent inverting buffers with tristate outputs (two banks of three, each with an independent active-low enable).

| /OE | Input | Output       |
|-----|-------|--------------|
| 0   | 0     | 1            |
| 0   | 1     | 0            |
| 1   | X     | Z (tristate) |

### Bank A (/OEa = GND, always enabled):

- Buffer 1: DO0 → /D0 (inverted DO0; feeds U8 Gates A and D)
- Buffer 2: DO1 → /D1 (inverted DO1; feeds U8 Gates B and C)
- Buffer 3: /CTL → CTL (active-high CTL strobe; drives U2 CS pin 34 and U10, U16 NAND inputs)
- Buffer 4: unused

### Bank B (/OEb = /SWE from U6, enabled only during status read):

- Buffer 5: TBMT (from U2 pin 22) → bDI7 = NOT(TBMT) (DI bus bit 7)
- Buffer 6: DR (from U2 pin 19) → bDI0 = NOT(DR) (DI bus bit 0)

## U3, U8, U10, U16 — 74LS00 (Quad 2-Input NAND Gate, DIP-14)

**Function:** Four 74LS00 packages, each containing four independent 2-input NAND gates, are used throughout the design:

- **U3** — 110-baud counter reload and the active-high RDE signal
- **U8** — set/clear inputs for the U9 interrupt-enable latches
- **U10** — interrupt qualification and combination
- **U16** — bus-strobe generation (/IO\_WR, /IO\_RD) and buffering the bus power-on-clear into the UART reset

### Truth Table (each gate)

| A | B | Y = NAND(A,B) |
|---|---|---------------|
| 0 | 0 | 1             |
| 0 | 1 | 1             |
| 1 | 0 | 1             |
| 1 | 1 | 0             |

### U3 — baud reload and RDE

**Gates A and B:** Unused; inputs tied off, outputs unconnected. (These gates drove the receive/transmit status LEDs in the prototype; the LEDs have been removed in this revision.)

**Gate C:** NAND(B110, U4 Q2) → U4 /PE (110-baud counter reload)

- When U4 Q3 (=B110) and Q2 are both high (count = 12–15): output = 0 → U4 loads preset = 2
- Implements the divide-by-11 feedback for 110-baud generation

**Gate D:** NAND(/RDE, +5V) = NOT(/RDE) → RDE (active-high data-read signal)

- Simply inverts the decoded active-low data-read signal to produce active-high RDE
- RDE drives U2 pin 16 (/SWE): when RDE=0 (not a data-port read), /SWE activates and U2 status outputs are enabled; when RDE=1 (data-port read), /SWE is inhibited to avoid conflict with RBR outputs

### U8 — interrupt latch set/clear

U8 generates the /S and /R inputs for U9 based on DO0, DO1, and the CTL strobe from U15 (confirmed from netlist):

- Gate A: NAND(CTL, /D0) → /CLR\_IN — drives U9 FF1 /R when CTL write and DO0=0
- Gate B: NAND(CTL, DO1) → /SET\_OUT — drives U9 FF2 /S when CTL write and DO1=1
- Gate C: NAND(CTL, /D1) → /CLR\_OUT — drives U9 FF2 /R when CTL write and DO1=0
- Gate D: NAND(CTL, DO0) → /SET\_IN — drives U9 FF1 /S when CTL write and DO0=1

### U10 — interrupt qualification

U10 qualifies the U9 interrupt-enable latches with the live UART status flags and combines them, producing the three interrupt-request signals routed to the jumper field (confirmed from netlist):

- Gate A: NAND(DR, Q1) → /IN\_INT — RX interrupt request: goes low when a character is ready (DR=1) **and** the input interrupt is enabled (U9 FF1 Q1=1)
- Gate B: NAND(TBMT, Q2) → /OUT\_INT — TX interrupt request: goes low when the transmitter can accept data (TBMT=1) **and** the output interrupt is enabled (U9 FF2 Q2=1)
- Gate C: NAND(/IN\_INT, /OUT\_INT) → intermediate
- Gate D: NAND(intermediate, +5V) = NOT(intermediate) → /BH\_INT — combined request: /BH\_INT = /IN\_INT AND /OUT\_INT, so it goes low when **either** /IN\_INT or /OUT\_INT is active

## U16 — bus strobes and power-on clear

U16 generates /IO\_WR and /IO\_RD from the processor status/control signals, and buffers the bus Power-On Clear (POC, Altair bus pin 99) into the UART master reset.

**Gate A:**  $\text{NAND}(/pWR, /pWR) = \text{NOT}(/pWR) \rightarrow pWR$  (intermediate)

- Inverts the processor active-low write strobe (/pWR, CPU control signal) to produce an active-high intermediate signal

**Gate B:**  $\text{NAND}(pWR, sOUT) \rightarrow /IO\_WR$

- /IO\_WR goes low (active) when a write cycle ( $pWR=1$ ) and the processor output-cycle status ( $sOUT=1$ ) are both asserted
- /IO\_WR drives U6 (74LS138) to decode I/O write operations

**Gate C:**  $\text{NAND}(/POC, /POC) = \text{NOT}(/POC) \rightarrow POC$

- Inverts the active-low bus power-on-clear signal (/POC, from Altair bus pin 99) to produce active-high POC
- POC drives U2 pin 21 (MR, Master Reset) high at power-up to reset the UART

**Gate D:**  $\text{NAND}(pDBIN, sINP) \rightarrow /IO\_RD$

- /IO\_RD goes low (active) when the processor data-bus-in strobe ( $pDBIN=1$ , CPU control signal) and input-cycle status ( $sINP=1$ ) are both asserted
- /IO\_RD drives U6 (74LS138) to decode I/O read operations

## U4 — 74LS161 (Synchronous 4-Bit Binary Counter, DIP-16)

**Function:** Synchronous 4-bit binary counter with synchronous parallel load and asynchronous clear. Generates the **B110** baud clock ( $\approx 1,745 \text{ Hz} \approx 16 \times 109 \text{ baud}$ ) by dividing the B1200 clock by approximately 11. The 110 baud rate cannot be obtained by binary division of the 1.2288 MHz oscillator, so this programmable divide-by-N counter is used.

| Pin | Name | Net         | Description                                                                |
|-----|------|-------------|----------------------------------------------------------------------------|
| 1   | /CLR | +5V         | Asynchronous clear disabled (tied high)                                    |
| 2   | CLK  | B1200       | Clock input: 19,200 Hz from U5 Counter B Q1 (pin 10)                       |
| 3   | D0   | GND         | Parallel data bit 0 = 0                                                    |
| 4   | D1   | +5V         | Parallel data bit 1 = 1                                                    |
| 5   | D2   | GND         | Parallel data bit 2 = 0                                                    |
| 6   | D3   | GND         | Parallel data bit 3 = 0 $\rightarrow$ preset value = 0b0010 = 2            |
| 7   | CEP  | +5V         | Count enable (always counting)                                             |
| 9   | /PE  | U3 Gate C   | Synchronous parallel load (active low); driven by NAND(Q3, Q2)             |
| 10  | CET  | +5V         | Count enable terminal (always counting)                                    |
| 11  | Q3   | B110        | MSB output $\rightarrow$ B110 $\approx 109 \text{ baud}$ 16 $\times$ clock |
| 12  | Q2   | Net-(U4-Q2) | Feedback to U3 Gate C for reload trigger                                   |
| 13  | Q1   | NC          | Unconnected                                                                |
| 14  | Q0   | NC          | Unconnected                                                                |

| Pin | Name | Net | Description                 |
|-----|------|-----|-----------------------------|
| 15  | TC   | NC  | Terminal count, unconnected |

**Divide-by-11 operation:** The counter starts at preset value 2 (loaded on reset or reload). It counts 2→3→4→...→12. When the count reaches 12, Q3=1 AND Q2=1, so U3 Gate C output = NAND(Q3, Q2) = 0 → /PE goes low → the counter synchronously loads preset value 2 on the next rising clock edge. This gives a cycle length of 11 clock periods.

$$19,200 \text{ Hz} \div 11 \approx 1,745 \text{ Hz} = \text{B110} (\approx 109 \text{ baud} \times 16)$$

The 0.8% error from ideal 110 baud is within RS-232 tolerance.

## U5 — 74LS393 (Dual 4-Bit Binary Counter, DIP-14)

**Function:** Two independent 4-bit ripple binary counters in a single package. Counter A is clocked directly by the 1.2288 MHz crystal oscillator (X1). Counter A's Q3 output (B4800) clocks Counter B. Together they generate seven of the eight available baud rate clocks (B300–B19200); U4 generates the eighth (B110).

### Baud Rate Outputs from U5 (1.2288 MHz input)

| Counter | Pin | Pinfunction | ÷ Factor | Frequency    | Net Name                                        |
|---------|-----|-------------|----------|--------------|-------------------------------------------------|
| A       | 1   | CP_1        | (input)  | 1,228,800 Hz | (oscillator)                                    |
| A       | 2   | MR_2        | reset    | —            | GND (never resets)                              |
| A       | 3   | Q0_3        | ÷2       | 614,400 Hz   | <b>Unconnected</b><br>(not used)                |
| A       | 4   | Q1_4        | ÷4       | 307,200 Hz   | <b>B19200</b> (16 × 19,200)                     |
| A       | 5   | Q2_5        | ÷8       | 153,600 Hz   | <b>B9600</b> (16 × 9,600)                       |
| A       | 6   | Q3_6        | ÷16      | 76,800 Hz    | <b>B4800</b> (16 × 4,800) →<br>clocks Counter B |
| B       | 13  | CP_13       | (input)  | 76,800 Hz    | (from Counter A Q3)                             |
| B       | 12  | MR_12       | reset    | —            | GND (never resets)                              |
| B       | 11  | Q0_11       | ÷2       | 38,400 Hz    | <b>B2400</b> (16 × 2,400)                       |
| B       | 10  | Q1_10       | ÷4       | 19,200 Hz    | <b>B1200</b> (16 × 1,200) →<br>clocks U4        |
| B       | 9   | Q2_9        | ÷8       | 9,600 Hz     | <b>B600</b> (16 × 600)                          |
| B       | 8   | Q3_8        | ÷16      | 4,800 Hz     | <b>B300</b> (16 × 300)                          |

## U11, U12 — 74LS85 (4-Bit Magnitude Comparator, DIP-16)

**Function:** Compares two 4-bit values (A and B) and asserts one of three outputs: A>B, A=B, or A<B. Two 74LS85 chips are cascaded to compare address bits **A1–A7** (7 bits) against the value programmed on SW3 (port address DIP switch). A0 is not compared here — it is decoded by U6 to distinguish the two port addresses. When A1–A7 match SW3, the board-select signal (BDSEL) is asserted, enabling the address decoder (U6).

**U11** compares the lower-order address bits **A1–A4** against **SA1–SA4**:

- A inputs: A3=A1, A2=A2, A1=A3, A0=A4
- B inputs: B3=SA1, B2=SA2, B1=SA3, B0=SA4
- Cascade inputs: Ia<b=GND, Ia=b=+5V, Ia>b=GND (initial “equal” seed)
- Oa=b output → U12 cascade input Ia=b

**U12** compares the upper-order address bits **A5–A7** against **SA5–SA7** (A3 bit position tied to GND = always-match):

- A inputs: A3=GND, A2=A5, A1=A6, A0=A7
- B inputs: B3=GND, B2=SA5, B1=SA6, B0=SA7
- Cascade input Ia=b: receives U11 Oa=b (previous bits all equal)
- Cascade inputs Ia<b=GND, Ia>b=GND
- **Oa=b output** → **BDSEL** (board select, active high)

### Truth Table (cascaded pair, relevant outputs only)

| A[7:1] vs. SW3 | Oa>b | Oa=b (BDSEL) | Oa<b |
|----------------|------|--------------|------|
| Address > SW3  | 1    | 0            | 0    |
| Address = SW3  | 0    | 1            | 0    |
| Address < SW3  | 0    | 0            | 1    |

Only Oa=b is used: when A1–A7 match SA1–SA7, BDSEL goes high and U6 is enabled.

## U13 — 74LS244 (Octal Tristate Buffer, DIP-20)

**Function:** Eight non-inverting tristate buffers in two banks of four, each with an independent active-low enable. Drives the Altair **DI bus** (data into the CPU) from the board's internal **bDI** bus.

Both banks (OEa and OEb) are controlled by **/DIEN** from U7 Gate A. /DIEN goes active (low) when either a status read or a data read is occurring, enabling all eight buffers simultaneously.

| /OE | Input | Output       |
|-----|-------|--------------|
| 0   | 0     | 0            |
| 0   | 1     | 1            |
| 1   | X     | Z (tristate) |

### Signal mapping (non-inverting — output = input):

| bDI Bus (input) | U13 Pin In   | U13 Pin Out  | Altair DI (output) |
|-----------------|--------------|--------------|--------------------|
| bDI0            | I0b (pin 11) | O0b (pin 9)  | DI0 = bDI0         |
| bDI1            | I3a (pin 8)  | O3a (pin 12) | DI1 = bDI1         |
| bDI2            | I1b (pin 13) | O1b (pin 7)  | DI2 = bDI2         |

| bDI Bus (input) | U13 Pin In   | U13 Pin Out  | Altair DI (output) |
|-----------------|--------------|--------------|--------------------|
| bDI3            | I2a (pin 6)  | O2a (pin 14) | DI3 = bDI3         |
| bDI4            | I2b (pin 15) | O2b (pin 5)  | DI4 = bDI4         |
| bDI5            | I1a (pin 4)  | O1a (pin 16) | DI5 = bDI5         |
| bDI6            | I3b (pin 17) | O3b (pin 3)  | DI6 = bDI6         |
| bDI7            | I0a (pin 2)  | O0a (pin 18) | DI7 = bDI7         |

**During status read (/SWE active, RDE=0):** All U2 status outputs are enabled. bDI0 = NOT(DR) and bDI7 = NOT(TBMT) via U15 Bank B; bDI2, bDI3, bDI4 driven directly by PE, FE, OVR (pins 13–15); remaining bits from U2 status word outputs on pins 5–12.

**During data read (/RDE active, RDE=1):** /SWE=1 — all U2 status outputs (DR, TBMT, PE, FE, OVR) are high-Z. bDI0–7 carry RBR receive data only.

## U14 — 74LS244 (Octal Tristate Buffer, DIP-20)

**Function:** Eight **non-inverting** tristate buffers. Passes the Altair **DO bus** (data output from the CPU) through to the board's internal **bDO bus** unchanged — output logic level equals input logic level. The bDO bus connects to U2 TBR1–TBR8 for transmit data.

Both bank enables (OEa and OEb) are tied to GND, so U14 is **permanently enabled** — the bDO bus continuously mirrors the Altair DO bus. This is harmless because the UART only latches the bDO bus when the /TDS data-write strobe pulses; no separate /DOEN gating is required.

**Signal mapping (non-inverting — output = input):**

| Altair DO (input) | U14 Pin In   | U14 Pin Out  | bDO Bus (output) |
|-------------------|--------------|--------------|------------------|
| DO0               | I0b (pin 11) | O0b (pin 9)  | bDO0 = DO0       |
| DO1               | I3a (pin 8)  | O3a (pin 12) | bDO1 = DO1       |
| DO2               | I1b (pin 13) | O1b (pin 7)  | bDO2 = DO2       |
| DO3               | I2a (pin 6)  | O2a (pin 14) | bDO3 = DO3       |
| DO4               | I2b (pin 15) | O2b (pin 5)  | bDO4 = DO4       |
| DO5               | I1a (pin 4)  | O1a (pin 16) | bDO5 = DO5       |
| DO6               | I3b (pin 17) | O3b (pin 3)  | bDO6 = DO6       |
| DO7               | I0a (pin 2)  | O0a (pin 18) | bDO7 = DO7       |

**Note:** DO0 and DO1 also connect **directly** (without going through U14) to U15 Bank A inputs. U15 Bank A **inverts** them to produce /D0 and /D1 for interrupt flip-flop control. U14 itself performs no inversion — the bDO bus is a buffered, logic-identical copy of the DO bus.

## U6 — 74LS138 (3-to-8 Line Decoder, DIP-16)

**Function:** Decodes three inputs into one of eight active-low outputs, subject to three enable inputs. Once the board is selected (BDSEL active), it decodes A0, /IO\_WR, and /IO\_RD to produce **four** qualified port-select signals — one for each of the four distinct operations (status read, data read, control write, data write).

**Pin connections:**

- Pin 1 (A0): A0 — distinguishes status/control port (A0=0) from data port (A0=1)
- Pin 2 (A1): /IO\_WR — active-low write strobe (HIGH = not writing; LOW = writing)
- Pin 3 (A2): /IO\_RD — active-low read strobe (HIGH = not reading; LOW = reading)
- Pin 4 (E1): GND — active-low enable, always enabled
- Pin 5 (E2): GND — active-low enable, always enabled
- Pin 6 (E3): BDSEL — active-high enable; only active when A1–A7 match SW3

**Port Decode Truth Table**

| BDSEL | /IO_RD | /IO_WR | A0 | Active Output   | Net Name | Function           |
|-------|--------|--------|----|-----------------|----------|--------------------|
| 1     | 0      | 1      | 0  | O2 (pin 13)     | /SWE     | Status read        |
| 1     | 0      | 1      | 1  | O3 (pin 12)     | /RDE     | Data read          |
| 1     | 1      | 0      | 0  | O4 (pin 11)     | /CTL     | Control write      |
| 1     | 1      | 0      | 1  | O5 (pin 10)     | /TDS     | Data write         |
| 0     | X      | X      | X  | None (all high) | —        | Board not selected |

Outputs O0, O1, O6, O7 are never asserted in normal operation (invalid or unused address combinations). When BDSEL is inactive, all outputs remain high (inactive).

**U9 — 74LS74 (Dual D Flip-Flop with Set and Reset, DIP-14)**

**Function:** Two independent positive-edge-triggered D flip-flops, each with asynchronous active-low set (/S) and reset (/R) inputs. Both D inputs and both CLK inputs are tied to GND, making the flip-flops operate in **asynchronous-only** mode: the Q output is controlled exclusively by /S (set) and /R (reset), never by the clock. The two Q outputs are the interrupt-enable latches; they feed U10, which qualifies them with the UART status flags and drives the Altair interrupt lines via JP3, JP4, and JP5.

**Truth Table (asynchronous control only)**

| /S | /R | Q | /Q |
|----|----|---|----|
| 0  | 1  | 1 | 0  |
| 1  | 0  | 0 | 1  |
| 0  | 0  | 1 | 1  |
| 1  | 1  | Q | /Q |

**FF1 — input (RX) interrupt enable:**

- **Set** (/S = /SET\_IN low): CTL write with DO0=1
- **Cleared** (/R = /CLR\_IN low): CTL write with DO0=0
- Q1 feeds U10 Gate A, where it is ANDed with DR to form the /IN\_INT request

**FF2 — output (TX) interrupt enable:**

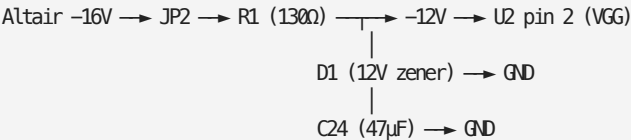
- **Set** (/S = /SET\_OUT low): CTL write with DO1=1
- **Cleared** (/R = /CLR\_OUT low): CTL write with DO1=0
- Q2 feeds U10 Gate B, where it is ANDed with TBMT to form the /OUT\_INT request

# Functional Operation

## Power Supply

The Altair bus provides +8V unregulated on pins 1 and 51. VR1 (LM7805, TO-220) regulates this to +5V for all logic ICs. The MAX232 (U1) uses this +5V with internal charge pumps to generate the RS-232  $\pm 10\text{V}$  levels — no external negative supply is needed for RS-232.

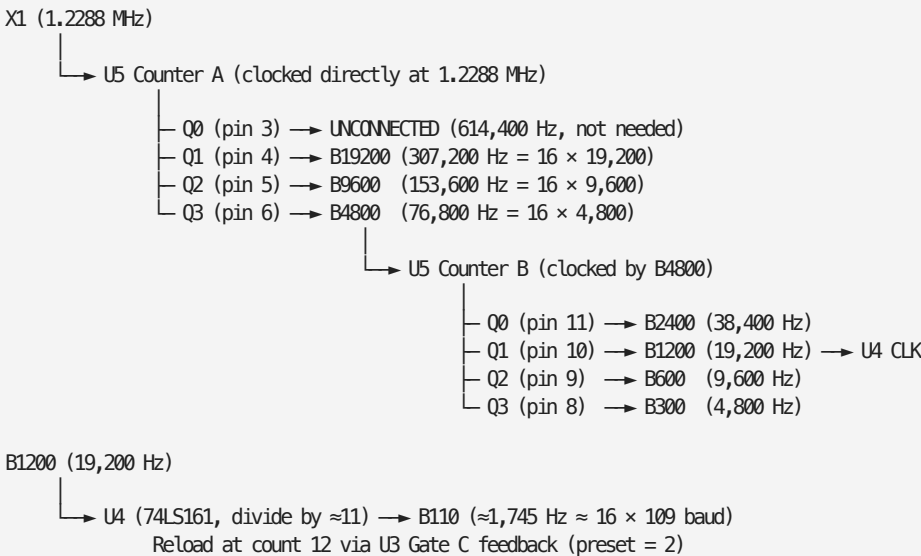
**−12V generation (UART VGG):** The COM2502 UART requires a −12V supply on its VGG pin (U2 pin 2). This is produced from the Altair −16V rail (bus pin 52) by a simple shunt (zener) regulator:



R1 drops the difference between −16V and −12V; D1, a 12V zener from the −12V node to ground, clamps the rail at −12V; C24 filters it. **JP2** is a 2-pin jumper in series with the −16V feed — it must be installed to enable the −12V supply. If a CMOS IM6402 is fitted instead of the COM2502, the −12V rail is not required and JP2 may be left open.

C21 (47µF) and C23 (47µF) provide bulk decoupling on the +8V input and +5V output of VR1, respectively.

## Baud Rate Generation



SW2 (8-position DIP switch) selects one of these eight baud rate clocks. The selected clock drives the **CLK** net, which connects to both RC (pin 17, Receiver Clock) and TC (pin 40, Transmitter Clock) of the UART (U2).

### SW2 Baud Rate Selection:

| SW2 Position | Baud Rate |
|--------------|-----------|
| 1            | 19,200    |
| 2            | 9,600     |
| 3            | 4,800     |

| SW2 Position | Baud Rate |
|--------------|-----------|
| 4            | 2,400     |
| 5            | 1,200     |
| 6            | 600       |
| 7            | 300       |
| 8            | 110       |

## Port Address Decode

The Altair bus carries an 8-bit I/O address (A0–A7) during IN and OUT instructions.

1. **U11, U12 (74LS85 cascade):** Compare address bits **A1–A7** against the value programmed on SW3 (7-position DIP switch). U11 compares A1–A4 vs SA1–SA4 (lower bits); U12 compares A5–A7 vs SA5–SA7 (upper bits). U11's equal output feeds U12's cascade input; U12's final A=B output asserts **BDSEL** (Board Select, active high). A0 is not part of this comparison.
2. **U6 (74LS138):** With BDSEL active as the G1 enable, U6 decodes A0, /IO\_WR, and /IO\_RD to produce exactly **four** port-select outputs:
  - **A0=0, reading:** /SWE (status read)
  - **A0=1, reading:** /RDE (data read)
  - **A0=0, writing:** /CTL (control write)
  - **A0=1, writing:** /TDS (data write)

## SW3 Port Address Configuration:

SW3 sets address bits A1–A7 for the board's base address (A0 is always 0 at the base). Switch positions map as follows:

| SW3 Position | Address Bit |
|--------------|-------------|
| 1 (pin 8)    | SA1 (A1)    |
| 2 (pin 9)    | SA2 (A2)    |
| 3 (pin 10)   | SA3 (A3)    |
| 4 (pin 11)   | SA4 (A4)    |
| 5 (pin 12)   | SA5 (A5)    |
| 6 (pin 13)   | SA6 (A6)    |
| 7 (pin 14)   | SA7 (A7)    |

SW3 selects any **even** base address from 0x00 to 0xFE (octal 000–376). The status/control port is always at the even address; the data port is at the next odd address (base + 1). A typical configuration with all switches closed places the card at port 0x00 (status/control at 0x00, data at 0x01), matching the original MITS Altair SIO.

## UART Format Configuration (SW1)

SW1 (5-position DIP switch) connects to U2 format control pins through resistor network RN1. A control port write (CS high) latches these settings into the UART:

| SW1 Position | UART Pin      | Function             |
|--------------|---------------|----------------------|
| 1            | NSB (pin 36)  | Stop bits (0=1, 1=2) |
| 2            | NDB1 (pin 38) | Word length bit 0    |

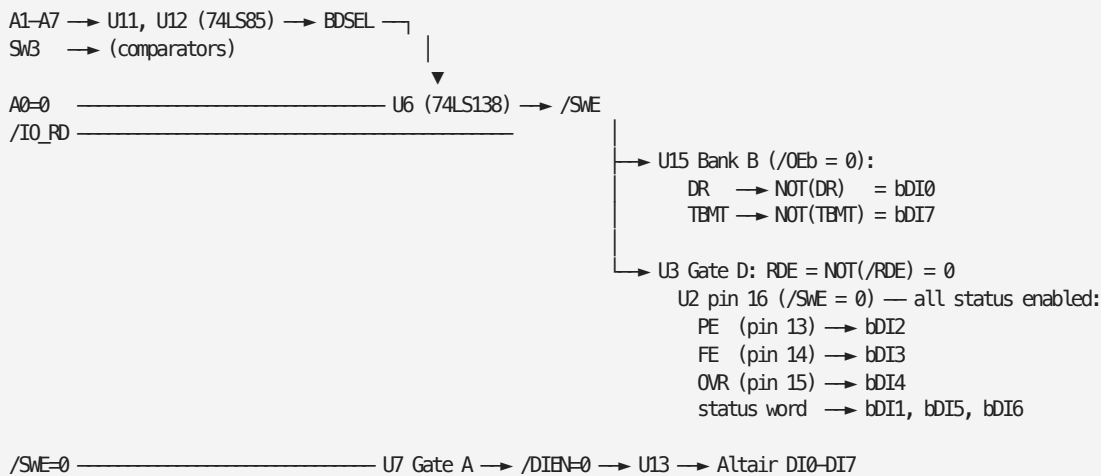
| SW1 Position | UART Pin      | Function               |
|--------------|---------------|------------------------|
| 3            | NDB2 (pin 37) | Word length bit 1      |
| 4            | POE (pin 39)  | Parity (0=even, 1=odd) |
| 5            | NPB (pin 35)  | No parity (1=disabled) |

## Bus Operation Flows

### Status Read

A **status read** occurs when the CPU executes an IN instruction to the status/control port address (A0=0).

Signal path:



1. CPU asserts the I/O address on A0–A7 and drives read strobes onto the Altair bus.
2. U11, U12 compare A1–A7 with SW3; if equal, **BDSEL** goes high.
3. U6 decodes A0=0 with read active → asserts **/SWE** (O2, active low).
4. /SWE enables **U15 Bank B**: NOT(DR) → bDI0; NOT(TBMT) → bDI7.
5. U3 Gate D derives **RDE=0** (not a data-port read) → U2 pin 16 (/SWE) = 0 → U2 enables all status outputs: status word on pins 5–12, PE (pin 13), FE (pin 14), OVR (pin 15), DR (pin 19), and TBMT (pin 22).
6. U7 Gate A: /DIEN = AND(/SWE, /RDE) → goes LOW → **U13** is enabled.
7. U13 drives bDI0–7 onto Altair **DI** bus (DI0–DI7).
8. CPU latches the status byte.

### Status byte bit assignments (on Altair DI bus):

| DI Bit | Source                           | Logic                                          |
|--------|----------------------------------|------------------------------------------------|
| 0      | U15 Bank B: NOT(DR)              | <b>0 = data ready</b> (DR high); 1 = no data   |
| 1      | U2 status word (bit unused)      | —                                              |
| 2      | U2 pin 13: PE (enabled by /SWE)  | 0 = no parity error; <b>1 = parity error</b>   |
| 3      | U2 pin 14: FE (enabled by /SWE)  | 0 = no framing error; <b>1 = framing error</b> |
| 4      | U2 pin 15: OVR (enabled by /SWE) | 0 = no overrun; <b>1 = overrun</b>             |

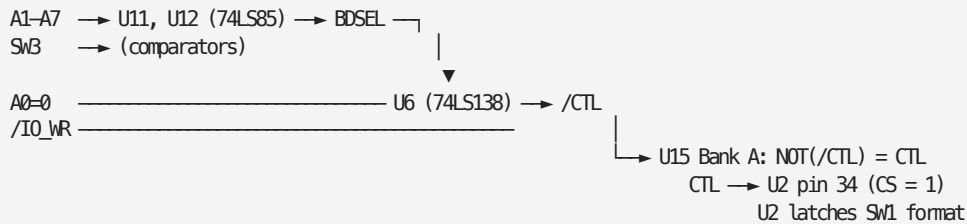
| DI Bit | Source                | Logic                                        |
|--------|-----------------------|----------------------------------------------|
| 5      | U2 status word (TBMT) | —                                            |
| 6      | U2 status word (DR)   | —                                            |
| 7      | U15 Bank B: NOT(TBMT) | <b>0 = TX ready</b> (TBMT high); 1 = TX busy |

Software polls bit 0 = 0 for receive ready, bit 7 = 0 for transmit ready, bits 2, 3, 4 for error conditions.

## Control Write

A **control write** occurs when the CPU executes an OUT instruction to the status/control port address (A0=0). This configures the UART word format and sets or clears the U9 flip-flop outputs that gate the interrupt drive circuitry.

Signal path:



(U14 is permanently enabled; the bD0 bus already mirrors Altair D00-D07)

```

Altair D00 --> U15 Bank A --> /D0 --> U8 Gate A: NAND(CTL, /D0) --> /CLR_IN --> U9 FF1 /R
Altair D00 --> U8 Gate D: NAND(CTL, D00) --> /SET_IN --> U9 FF1 /S
Altair D01 --> U15 Bank A --> /D1 --> U8 Gate C: NAND(CTL, /D1) --> /CLR_OUT --> U9 FF2 /R
Altair D01 --> U8 Gate B: NAND(CTL, D01) --> /SET_OUT --> U9 FF2 /S
  
```

1. CPU places the control word on the Altair **DO** bus and drives write strobes.
2. U6 decodes A0=0 with write → **/CTL** active (O4).
3. U15 Bank A, buffer 3: inverts **/CTL** → **CTL** (active high) → U2 pin 34 (CS).
4. U2 CS high → UART latches the format control bits from SW1 pins (NDB1, NDB2, NSB, NPB, POE).

**Simultaneously**, DO0 and DO1 gate U8 NAND outputs to drive /S and /R on U9:

| DO bus | Signal      | NAND Gate | Output                   |
|--------|-------------|-----------|--------------------------|
| D0 = 1 | CTL AND DO0 | U8 Gate D | /SET_IN low → U9 FF1 /S  |
| D0 = 0 | CTL AND /D0 | U8 Gate A | /CLR_IN low → U9 FF1 /R  |
| D1 = 1 | CTL AND DO1 | U8 Gate B | /SET_OUT low → U9 FF2 /S |
| D1 = 0 | CTL AND /D1 | U8 Gate C | /CLR_OUT low → U9 FF2 /R |

**Interrupt enable/disable (DO0 and DO1):**

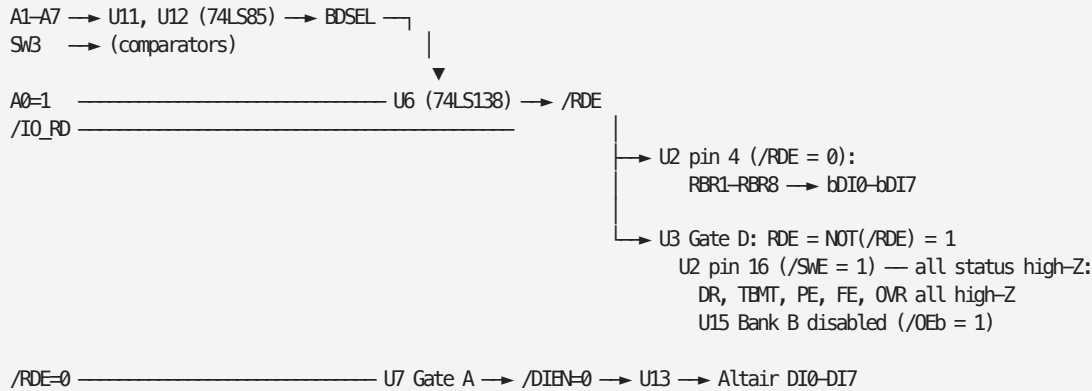
| DO0 | DO1 | Input (RX) Interrupt | Output (TX) Interrupt |
|-----|-----|----------------------|-----------------------|
| 0   | 0   | Disabled             | Disabled              |
| 1   | 0   | Enabled              | Disabled              |
| 0   | 1   | Disabled             | Enabled               |
| 1   | 1   | Enabled              | Enabled               |

To enable the input interrupt and disable the output interrupt, write 0x01 (D0=1, D1=0) to the control port. Bits D2–D7 are don't-care for interrupt control.

## Data Read

A **data read** occurs when the CPU executes an IN instruction to the data port (A0=1), after confirming bit 0 = 0 in the status byte.

Signal path:



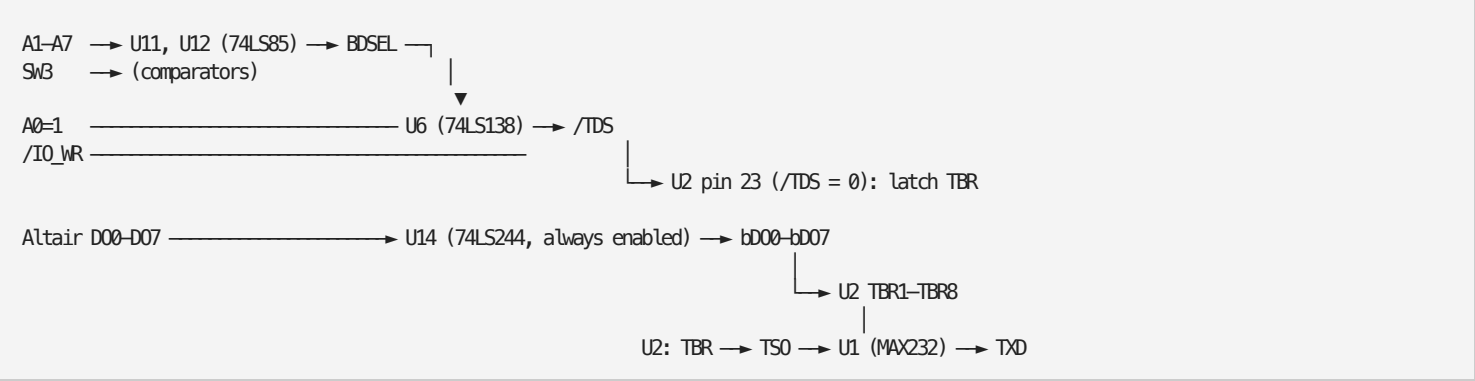
1. CPU asserts the data port address on A0–A7 (A0=1) and drives read strobes.
2. U11, U12 compare and assert BDSEL; U6 decodes A0=1 with read → **/RDE** active (O3).
3. /RDE asserts U2 pin 4 (/RDE) → U2 enables RBR1–RBR8 outputs onto bDI0–7.
4. U3 Gate D:  $RDE = \text{NOT}(/RDE) = 1$  → U2 pin 16 (/SWE) = 1 → all U2 status outputs (DR, TBMT, PE, FE, OVR) go high-Z.
5. U15 Bank B is NOT enabled (/OEB = /SWE = 1).
6. U7 Gate A: /DIEN goes LOW → U13 enabled → drives bDI0–7 (= RBR1–8) onto Altair DI bus.
7. CPU latches the received character.
8. U2's DR flag is cleared automatically when /RDAV (tied to DR) is pulled low at the end of the cycle.

**Note:** During a data read (RDE=1, /SWE=1), all U2 status outputs are high-Z. bDI0–7 carries RBR receive data only.

## Data Write (Transmit)

A **data write** occurs when the CPU executes an OUT instruction to the data port (A0=1) to transmit a character.

Signal path:

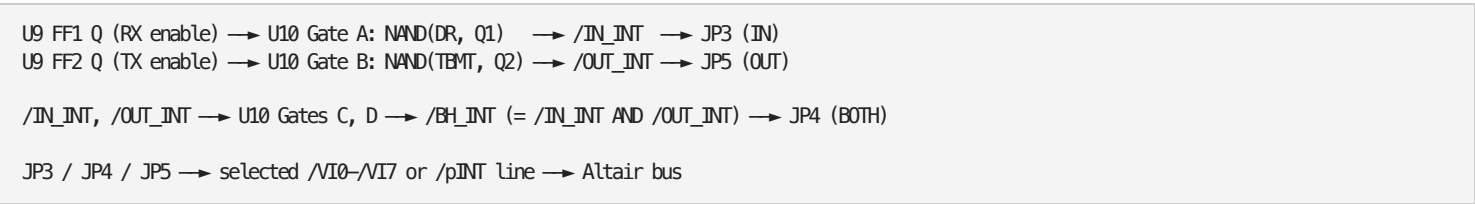


1. CPU places the character on the Altair **DO** bus (DO0–DO7) and drives write strobes.
2. U11, U12 compare and assert BDSEL; U6 decodes A0=1 with write → **/TDS** active (O5).
3. U14 (permanently enabled) already mirrors DO0–DO7 onto bDO0–7, which connects to U2 TBR1–TBR8.
4. /TDS asserts U2 pin 23 (/TDS) directly → U2 latches TBR from the bDO bus.
5. U2 transfers the character to the transmit shift register and serially transmits via **TSO** at the programmed baud rate.
6. U1 (MAX232) converts TSO (TTL) to RS-232 levels on the **TXD** line.
7. When TBR is emptied, **TBMT** returns high; status bit 7 returns to 0 (TX ready).

Interrupts

The SIOA board drives /pINT or a selected /VI line on the Altair bus active-low. The interrupt requests are produced by U10, which qualifies the U9 interrupt-enable latches with the live UART status flags. Each request is then routed to a target bus line through one of three jumper headers (JP3, JP4, JP5).

Signal Path



- **/IN\_INT** asserts (low) when a received character is ready (DR=1) and the RX interrupt is enabled.
- **/OUT\_INT** asserts (low) when the transmitter is ready (TBMT=1) and the TX interrupt is enabled.
- **/BH\_INT** is the combined request — it asserts when either /IN\_INT or /OUT\_INT is active.

Jumper Selection

Each jumper is a 2×9 header. One column carries the interrupt-request signal; the other column exposes the nine possible targets — /VI0 through /VI7 and /pINT (pins 17–18). A shunt connects the request to exactly one target.

| Jumper     | Request  | Altair Line Driven                          |
|------------|----------|---------------------------------------------|
| JP3 (IN)   | /IN_INT  | one of /VI0–/VI7 or /pINT (RX source)       |
| JP4 (BOTH) | /BH_INT  | one of /VI0–/VI7 or /pINT (combined source) |
| JP5 (OUT)  | /OUT_INT | one of /VI0–/VI7 or /pINT (TX source)       |

## Priority Assignment

The /VI lines connect to the 88-VI Vectored Interrupt Board. Priority levels 0–7 correspond to /VI0–/VI7, with **0 being the lowest priority and 7 the highest**.

The three jumpers allow two interrupt priority configurations:

- **Separate priorities** — Install a shunt on JP3 (IN) and a separate shunt on JP5 (OUT). The receive and transmit interrupts are routed to different /VI lines, giving each a distinct priority level on the 88-VI interrupt controller.
- **Single shared priority** — Install a single shunt on JP4 (BOTH). The receive and transmit interrupts are combined onto one /VI line and share a single priority level.

## Non-Vectored Interrupt (/pINT)

If the 88-VI vectored interrupt board is not present, place the shunt on the **/pINT** position (pins 17–18) of the desired jumper — JP3 (IN), JP4 (BOTH), or JP5 (OUT) — to drive the processor's interrupt line (/pINT, Altair pin 73) directly. This generates a single non-vectored interrupt to the CPU.

When using /PINT, the processor jumps immediately to octal location 70 (0x38) on interrupt. Place the interrupt service routine in locations **70–77 octal (0x38–0x3F)**.

## Power-On Clear (POC)

U16 Gate C:  $\text{NAND}(/POC, /POC) = \text{NOT}(/POC) \rightarrow \text{POC}$  (active high) drives U2 pin 21 (MR, Master Reset) high briefly at power-up to:

- Clear all UART status flags
- Reset the receiver and transmitter
- Ensure a known state before software initialization

The /POC signal is the Altair bus Power-On Clear line (bus pin 99), driven low briefly at power-up by the host system (front panel / power-on reset circuitry). The board receives it on pin 99 and uses it to reset the UART; it is not generated on the card.

## Connector Summary

| Ref | Description | Connector Type | Notes                                                                                                                                                                                                      |
|-----|-------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| J1  | TTL Header  | 1×6 pin header | TTL-level TX (TSO, pin 5) and RX (RSI, pin 4), no level shifting; pins 1–2 GND. Pinout matches the standard FTDI USB-to-serial 6-pin TTL header, so an FTDI USB-serial cable/adaptor can connect directly. |
| J2  | DE9 DCE     | 2×5 IDC header | 9-pin DCE RS-232 (DTE device connects here)                                                                                                                                                                |

| Ref | Description | Connector Type | Notes                                        |
|-----|-------------|----------------|----------------------------------------------|
| J3  | DB25 DCE    | 2×7 IDC header | 25-pin DCE RS-232 (DTE device connects here) |
| J4  | DE9 DTE     | 2×5 IDC header | 9-pin DTE RS-232 (DCE device connects here)  |
| J5  | DB25 DTE    | 2×7 IDC header | 25-pin DTE RS-232 (DCE device connects here) |

## RS-232 Connector Wiring

The four RS-232 ports are 2-row IDC pin headers, not D-sub connectors. The schematic annotation **CROSSOVER / DTK / INTEL / SUPERMICRO** does **not** describe a null-modem cable — it names the **ribbon-cable pin-mapping convention** used between a 2-row IDC header and a D-sub connector. The board's header pins are arranged so that a flat ribbon cable terminated with an **IDC-to-D-sub crimp connector** produces a correct, standard RS-232 D-sub pinout. This is what lets the board be cabled out with ordinary crimp-on connectors instead of soldered D-sub.

In this convention the ribbon conductors alternate between the two rows of D-sub pins: odd IDC pins map to the low-numbered D-sub pins, even IDC pins to the high-numbered ones.

### Crimp pin mapping (IDC header → D-sub):

| IDC pin | DE9 pin (2×5) | DB25 pin (2×7) |
|---------|---------------|----------------|
| 1       | 1             | 1              |
| 2       | 6             | 14             |
| 3       | 2             | 2              |
| 4       | 7             | 15             |
| 5       | 3             | 3              |
| 6       | 8             | 16             |
| 7       | 4             | 4              |
| 8       | 9             | 17             |
| 9       | 5             | 5              |
| 10      | — (key)       | 18             |
| 11      | —             | 6              |
| 12      | —             | 19             |
| 13      | —             | 7              |
| 14      | —             | 20             |

**Connector gender:** the **DTE** headers (J4, J5) terminate to a **male** D-sub; the **DCE** headers (J2, J3) terminate to a **female** D-sub. Only TXD, RXD, and signal ground are wired — no hardware-handshake lines are connected. Signal direction is from the board's point of view: **TXD = board transmit (output)**, **RXD = board receive (input)**.

Note that DE9 and DB25 historically swap the TXD/RXD pin assignments (on a DTE, TXD is DE9 pin 3 but DB25 pin 2), and DTE vs. DCE swaps them again. The header wiring below already accounts for both, so every resulting D-sub is a standard, straight-through-cable-compatible pinout.

**DCE headers — female D-sub (J2, J3)**

A DTE device (terminal/PC) connects here with a straight-through cable.

**J2 — DE9 DCE (female):**

| IDC pin | Signal          | DE9 pin |
|---------|-----------------|---------|
| 3       | TXD (board out) | 2       |
| 5       | RXD (board in)  | 3       |
| 9       | GND             | 5       |

**J3 — DB25 DCE (female):**

| IDC pin | Signal          | DB25 pin |
|---------|-----------------|----------|
| 3       | RXD (board in)  | 2        |
| 5       | TXD (board out) | 3        |
| 13      | GND             | 7        |

**DTE headers — male D-sub (J4, J5)**

A DCE device (modem) connects here with a straight-through cable.

**J4 — DE9 DTE (male):**

| IDC pin | Signal          | DE9 pin |
|---------|-----------------|---------|
| 3       | RXD (board in)  | 2       |
| 5       | TXD (board out) | 3       |
| 9       | GND             | 5       |

**J5 — DB25 DTE (male):**

| IDC pin | Signal          | DB25 pin |
|---------|-----------------|----------|
| 3       | TXD (board out) | 2        |
| 5       | RXD (board in)  | 3        |
| 13      | GND             | 7        |

## Jumper Summary

| Ref | Label | Type       | Function                                                                                                                                                       |
|-----|-------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JP1 | 5V    | 1×2 header | Routes +5V onto the DE9 DCE connector (J2) for powering an external device. Leave open if not needed.                                                          |
| JP2 | -12V  | 1×2 header | Connects the Altair -16V rail to the on-board -12V regulator. <b>Install when using a COM2502</b> (which needs -12V on VGG); leave open for a +5V-only IM6402. |

| Ref | Label | Type       | Function                                                                     |
|-----|-------|------------|------------------------------------------------------------------------------|
| JP3 | IN    | 2×9 header | Routes the RX interrupt request (/IN_INT) to one of /VIO–/V17 or /pINT       |
| JP4 | BOTH  | 2×9 header | Routes the combined interrupt request (/BH_INT) to one of /VIO–/V17 or /pINT |
| JP5 | OUT   | 2×9 header | Routes the TX interrupt request (/OUT_INT) to one of /VIO–/V17 or /pINT      |

## DIP Switch Summary

| Switch | Positions | Function                                                                                      |
|--------|-----------|-----------------------------------------------------------------------------------------------|
| SW1    | 5         | UART word format: NDB1, NDB2, NSB, NPB, POE (word length, stop bits, parity)                  |
| SW2    | 8         | Baud rate select: 110 / 300 / 600 / 1200 / 2400 / 4800 / 9600 / 19200                         |
| SW3    | 7         | I/O port base address: bits A1–A7 (switch 1=A1, switch 7=A7); A0 selects status vs. data port |

## Key Internal Signal Reference

| Signal | Source         | Destination                             | Description                               |
|--------|----------------|-----------------------------------------|-------------------------------------------|
| CLK    | SW2 output     | U2 RC, U2 TC                            | Selected baud rate clock (16× baud)       |
| RSI    | U1 R1OUT       | U2 pin 20                               | TTL receive data from MAX232              |
| TSO    | U2 pin 25      | U1 T1IN                                 | TTL transmit data to MAX232               |
| TXD    | U1 T1OUT       | J2, J3, J4, J5                          | RS-232 transmit line                      |
| RXD    | J2, J3, J4, J5 | U1 R1IN                                 | RS-232 receive line                       |
| TBMT   | U2 pin 22      | U15 Bank B (pin 12), U10 Gate B (pin 4) | TX buffer empty flag                      |
| DR     | U2 pin 19      | U15 Bank B (pin 14), U10 Gate A (pin 1) | Received data ready flag                  |
| /SWE   | U6 O2 (pin 13) | U15 OEb, U7 Gate A                      | Decoded status read strobe (active low)   |
| /RDE   | U6 O3 (pin 12) | U2 pin 4, U7 Gate A, U3 Gate D          | Decoded data read strobe (active low)     |
| RDE    | U3 Gate D      | U2 pin 16 (/SWE)                        | Active-high data read signal = NOT(/RDE)  |
| /CTL   | U6 O4 (pin 11) | U15 Bank A I3                           | Decoded control write strobe (active low) |

| Signal      | Source                     | Destination                    | Description                                                            |
|-------------|----------------------------|--------------------------------|------------------------------------------------------------------------|
| /TDS        | U6 O5 (pin 10)             | U2 pin 23                      | Decoded data write strobe (active low)                                 |
| CTL         | U15 Bank A O3 (pin 7)      | U2 pin 34 (CS), U8             | Active-high control strobe                                             |
| /pWR        | CPU (bus pin 77)           | U16 Gate A inputs (pins 1, 2)  | Processor write strobe (active low); CPU control signal                |
| pWR         | U16 Gate A (pin 3)         | U16 Gate B input (pin 4)       | Intermediate active-high write signal = NOT(/pWR)                      |
| sOUT        | CPU (bus pin 45)           | U16 Gate B input (pin 5)       | Processor output-cycle status (active high); CPU status signal         |
| /IO_WR      | U16 Gate B (pin 6)         | U6 pin 2                       | I/O write strobe: low when pWR and sOUT both active                    |
| pDBIN       | CPU (bus pin 78)           | U16 Gate D input (pin 12)      | Processor data-bus-in strobe (active high); CPU control signal         |
| sINP        | CPU (bus pin 46)           | U16 Gate D input (pin 13)      | Processor input-cycle status (active high); CPU status signal          |
| /IO_RD      | U16 Gate D (pin 11)        | U6 pin 3                       | I/O read strobe: low when pDBIN and sINP both active                   |
| /POC        | Altair bus (pin 99)        | U16 Gate C inputs (pins 9, 10) | Bus power-on-clear (active low); system reset at power-up              |
| /D0         | U15 Bank A O1 (pin 3)      | U8 Gates A, D                  | NOT(DO0)                                                               |
| /D1         | U15 Bank A O2 (pin 5)      | U8 Gates B, C                  | NOT(DO1)                                                               |
| /DIEN       | U7 Gate A (pin 3)          | U13 OEa, OEb                   | DI bus enable: low during any read                                     |
| BDSEL       | U12 Oa=b (pin 6)           | U6 E3 (pin 6)                  | Board address match (active high)                                      |
| bDI0–7      | U2 RBR, status, U15 Bank B | U13 inputs                     | Board-side DI bus                                                      |
| bDO0–7      | U14 outputs                | U2 TBR1–TBR8                   | Board-side DO bus                                                      |
| B110–B19200 | U4, U5 outputs             | SW2 inputs                     | Available baud rate clocks                                             |
| POC         | U16 Gate C output (pin 8)  | U2 pin 21 (MR)                 | Active-high master reset; NOT(/POC)                                    |
| Q1          | U9 FF1 Q (pin 5)           | U10 Gate A (pin 2)             | RX interrupt enable latch                                              |
| Q2          | U9 FF2 Q (pin 9)           | U10 Gate B (pin 5)             | TX interrupt enable latch                                              |
| /IN_INT     | U10 Gate A (pin 3)         | JP3, U10 Gate C                | RX interrupt request: low when DR and Q1 both high                     |
| /OUT_INT    | U10 Gate B (pin 6)         | JP5, U10 Gate C                | TX interrupt request: low when TBMT and Q2 both high                   |
| /BH_INT     | U10 Gate D (pin 11)        | JP4                            | Combined interrupt request: low when either /IN_INT or /OUT_INT active |
| /VI0–VI7    | JP3, JP4, JP5              | Altair bus (pins 4–11)         | Vectored interrupt outputs (active low)                                |

| Signal | Source              | Destination             | Description                                         |
|--------|---------------------|-------------------------|-----------------------------------------------------|
| /pINT  | JP3, JP4, JP5       | CPU (Altair bus pin 73) | Processor interrupt request (active low)            |
| -16V   | Altair bus (pin 52) | JP2 → R1                | Unregulated negative rail for -12V generation       |
| -12V   | D1/R1/C24 node      | U2 pin 2 (VGG)          | On-board -12V supply for UART (12V zener regulated) |
| R1     | JP2-B node          | -12V node               | 130Ω series resistor in the -12V zener regulator    |
| D1     | -12V node           | GND                     | 12V zener; clamps the -12V rail                     |

## Bill of Materials

| Ref                                                                        | Qty | Value           | Footprint                                    |
|----------------------------------------------------------------------------|-----|-----------------|----------------------------------------------|
| C1, C2, C3, C5, C6                                                         | 5   | 1uF             | Disc capacitor, D4.3mm, P5.00mm              |
| C4, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C22 | 16  | 0.1uF           | Disc capacitor, D4.3mm, P5.00mm              |
| C21, C23, C24                                                              | 3   | 47uF            | Tantalum radial, D6.0mm, P2.50mm             |
| D1                                                                         | 1   | 12V             | Axial diode, DO-15, P12.70mm                 |
| HS1                                                                        | 1   | Heatsink        | Aavid TV5G, TO-220 horizontal                |
| J1                                                                         | 1   | TTL HEADER      | Pin header 1×6, P2.54mm vertical             |
| J2                                                                         | 1   | DE9 DCE HEADER  | IDC header 2×5, P2.54mm vertical             |
| J3                                                                         | 1   | DB25 DCE HEADER | IDC header 2×7, P2.54mm vertical             |
| J4                                                                         | 1   | DE9 DTE HEADER  | IDC header 2×5, P2.54mm vertical             |
| J5                                                                         | 1   | DB25 DTE HEADER | IDC header 2×7, P2.54mm vertical             |
| JP1                                                                        | 1   | 5V              | Pin header 1×2, P2.54mm vertical             |
| JP2                                                                        | 1   | -12V            | Pin header 1×2, P2.54mm vertical             |
| JP3                                                                        | 1   | IN              | Pin header 2×9, P2.54mm vertical             |
| JP4                                                                        | 1   | BOTH            | Pin header 2×9, P2.54mm vertical             |
| JP5                                                                        | 1   | OUT             | Pin header 2×9, P2.54mm vertical             |
| R1                                                                         | 1   | 130             | Axial resistor, L9.9mm D3.6mm, P12.70mm      |
| RN1                                                                        | 1   | 4.7K            | Resistor array SIP-6                         |
| RN2                                                                        | 1   | 2.2K            | Resistor array SIP-8                         |
| SW1                                                                        | 1   | SW_DIP_x05      | DIP switch 5-position, 9.78×14.88mm, P2.54mm |
| SW2                                                                        | 1   | SW_DIP_x08      | DIP switch 8-position, 9.78×22.5mm, P2.54mm  |

| Ref              | Qty | Value      | Footprint                                    |
|------------------|-----|------------|----------------------------------------------|
| SW3              | 1   | SW_DIP_x07 | DIP switch 7-position, 9.78×19.96mm, P2.54mm |
| U1               | 1   | MAX232     | DIP-16 socket, W7.62mm                       |
| U2               | 1   | COM2502    | DIP-40 socket, W15.24mm                      |
| U3, U8, U10, U16 | 4   | 74LS00     | DIP-14 socket, W7.62mm                       |
| U4               | 1   | 74LS161    | DIP-16 socket, W7.62mm                       |
| U5               | 1   | 74LS393    | DIP-14 socket, W7.62mm                       |
| U6               | 1   | 74LS138    | DIP-16 socket, W7.62mm                       |
| U7               | 1   | 74LS08     | DIP-14 socket, W7.62mm                       |
| U9               | 1   | 74LS74     | DIP-14 socket, W7.62mm                       |
| U11, U12         | 2   | 74LS85     | DIP-16 socket, W7.62mm                       |
| U13, U14         | 2   | 74LS244    | DIP-20 socket, W7.62mm                       |
| U15              | 1   | 74LS368    | DIP-16 socket, W7.62mm                       |
| VR1              | 1   | LM7805     | TO-220-3 horizontal, tab down                |
| X1               | 1   | 1.2288 MHz | Oscillator DIP-8                             |